

DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 5 (AT2)

Visualization Development Exercise

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

Question 1 — Chart Selection (Topic 1: Directory of Visualizations)

Task: For each scenario below, name the single best chart type from the directory (Amounts / Distribution / Proportion / x–y Relationship / Geospatial / Network / Temporal / Multivariate) and justify your choice in 1–2 sentences. No code required for this question.

1. Comparing total revenue across 6 product categories in one year.
2. Showing how the salary distribution differs across 4 job levels.
3. Showing market share of 5 competing brands.
4. Showing the relationship between advertising spend and sales revenue.
5. Showing monthly website traffic over 3 years.
6. Showing which employees report to which managers in an org.

ANSWER:

1. Comparing total revenue across 6 product categories in one year

Chart Type: Bar Chart (Amounts)

Justification:

A bar chart is the best choice because it compares values across different categories. It makes it easy to identify which product category has the highest and lowest revenue.

2. Showing how the salary distribution differs across 4 job levels

Chart Type: Violin Plot (Distribution)

Justification:

A violin plot displays the distribution, spread, and density of salary values for each job level. It helps compare how salaries vary across different levels.

3. Showing market share of 5 competing brands

Chart Type: Pie Chart (Proportion)

Justification:

A pie chart clearly shows the proportion of total market share held by each brand. It helps visualize percentage contributions effectively.

4. Showing the relationship between advertising spend and sales revenue

Chart Type: Scatter Plot (x–y Relationship)

Justification:

A scatter plot is ideal for showing the relationship between two continuous variables. It helps identify positive, negative, or no correlation.

5. Showing monthly website traffic over 3 years

Chart Type: Line Chart (Temporal)

Justification:

A line chart is best for time-series data because it clearly shows trends, seasonal variations, and changes over time.

6. Showing which employees report to which managers in an organization

Chart Type: Network Diagram (Organizational Chart)

Justification:

A network diagram visually represents reporting relationships between employees and managers using connected nodes.

Question 2

Dataset — avg_score.csv

Subject	Average_Score
Mathematics	72
Science	68
English	75
History	64
Computer Sci	81

Task:

- Create a vertical bar plot of Average_Score by Subject, sorted in descending order of score.
- Color the bars using a single consistent color; highlight the highest-scoring subject in a different color.
- Add data labels on top of each bar.

Python: use pandas to build the DataFrame and matplotlib.pyplot.bar (or seaborn.barplot). **R:** use a data.frame and ggplot2::geom_col().

ANSWER:**Python Code:**

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'Subject': ['Mathematics', 'Science', 'English', 'History', 'Computer Sci'],
    'Average_Score': [72, 68, 75, 64, 81]
})

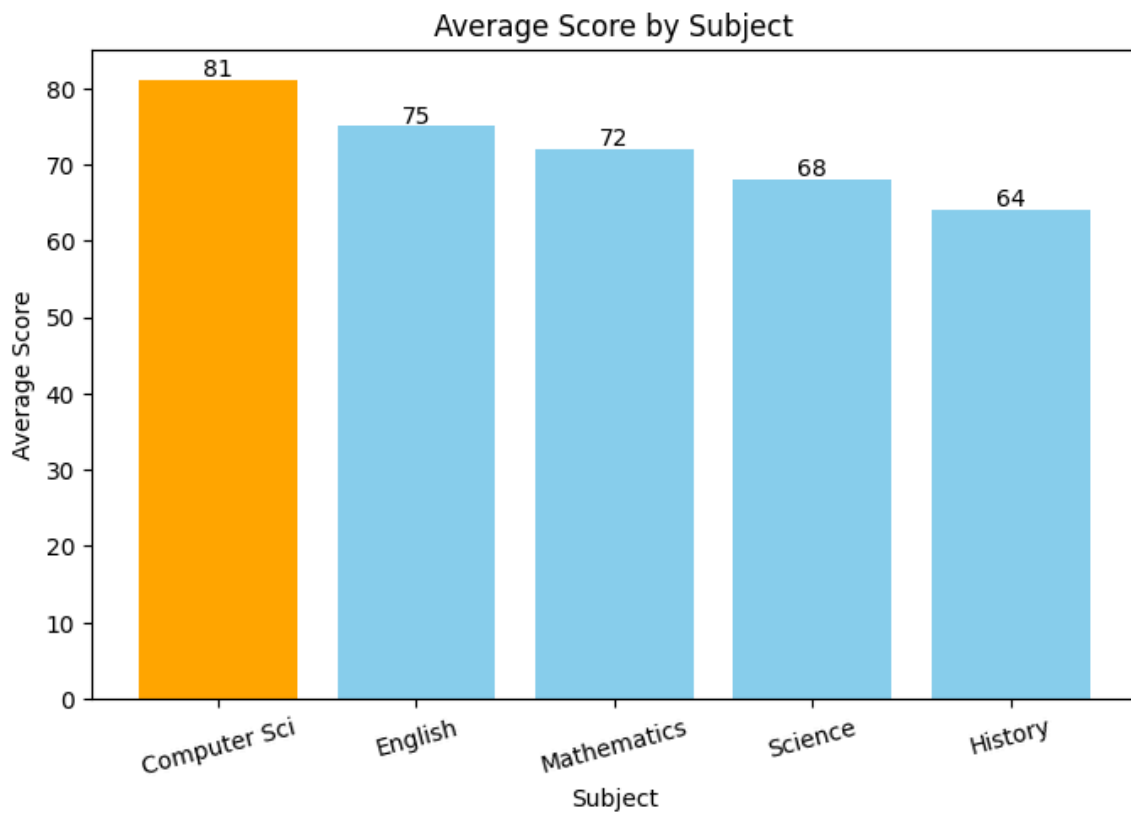
df = df.sort_values(by='Average_Score', ascending=False)
colors = ['orange' if score == df['Average_Score'].max() else 'skyblue'
          for score in df['Average_Score']]

plt.figure(figsize=(8,5))
bars = plt.bar(df['Subject'], df['Average_Score'], color=colors)
for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2,
             bar.get_height() + 0.5,
             str(int(bar.get_height())),
             ha='center',
             fontsize=10)

plt.title("Average Score by Subject")
plt.xlabel("Subject")
plt.ylabel("Average Score")
plt.xticks(rotation=15)
```

```
plt.show()
```

Output:



R Code:

```
library(ggplot2)
df <- data.frame(
  Subject = c("Mathematics",
              "Science",
              "English",
              "History",
              "Computer Sci"),
  Average_Score = c(72,68,75,64,81)
)
df <- df[order(-df$Average_Score), ]
```

```

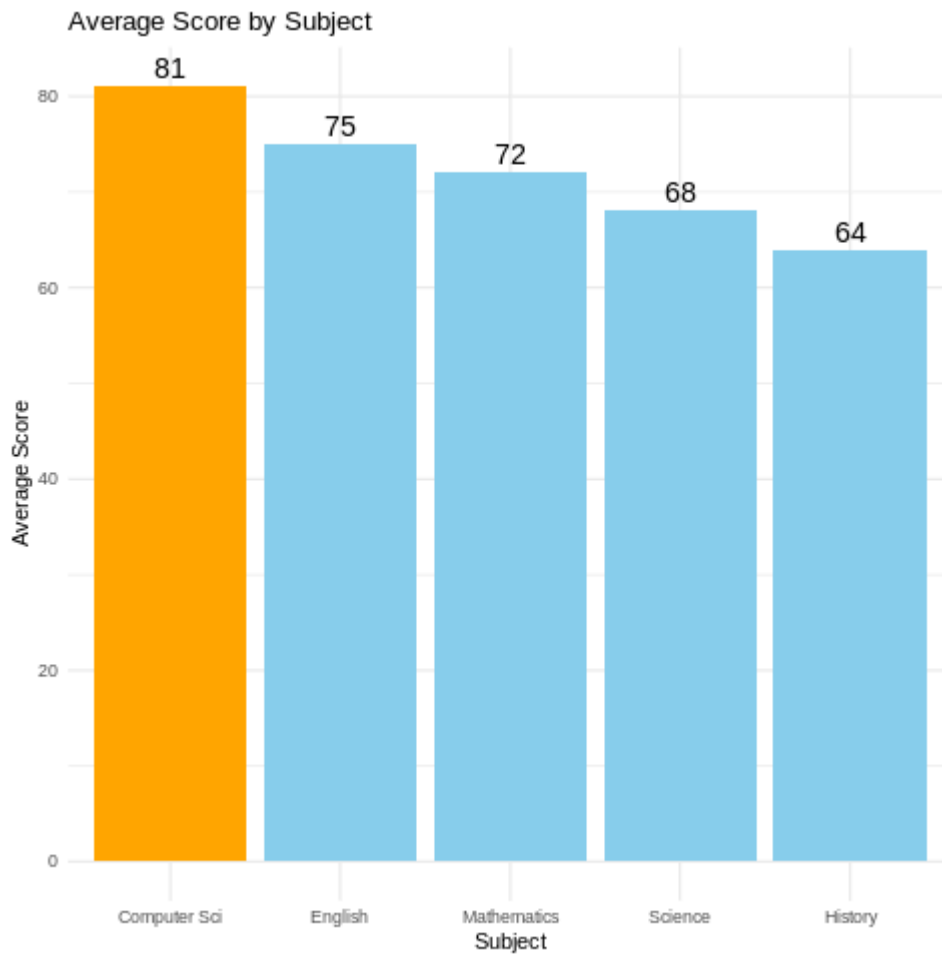
df$Color <- ifelse(df$Average_Score == max(df$Average_Score),
  "Highest",
  "Others")

ggplot(df,
  aes(x = reorder(Subject, -Average_Score),
    y = Average_Score,
    fill = Color)) +

  geom_col() +
  geom_text(aes(label = Average_Score),
    vjust = -0.4,
    size = 5) +
  scale_fill_manual(values = c("Highest" = "orange",
    "Others" = "skyblue")) +
  labs(title = "Average Score by Subject",
    x = "Subject",
    y = "Average Score") + theme_minimal() + theme(legend.position =
  "none")

```

Output:



Interpretation:

- Computer Science has the highest average score (81) and is highlighted in orange.
- History has the lowest average score (64).
- The descending order makes it easy to compare the performance of all subjects.
- Overall, students performed best in Computer Science and English, while History had the lowest average score.

Question 3

Dataset — regional_sales.csv

Region	Quarter	Sales_000s
North	Q1	120
North	Q2	135
North	Q3	128
North	Q4	150
South	Q1	95
South	Q2	110
South	Q3	105
South	Q4	130
East	Q1	140
East	Q2	138
East	Q3	145

East	Q4	160
West	Q1	80
West	Q2	85
West	Q3	90
West	Q4	100

Task (produce all four views from the same dataset):

1. Grouped bar chart: Sales_000s by Region, grouped by Quarter.
2. Stacked bar chart: same data, stacked by Quarter within each Region.
3. Dot plot: total annual sales per region (sum across quarters), ranked from

highest to lowest. 4. Heatmap: Region (rows) $\times$ Quarter (columns), color-encoded by Sales_000s.

Python: pandas.pivot_table to reshape; matplotlib/seaborn for grouped/stacked bars, seaborn.heatmap for the heatmap, matplotlib.pyplot.scatter (or plt.plot with markers) for the dot plot.

R: tidyr::pivot_wider where needed; ggplot2::geom_col(position = "dodge") for grouped, position = "stack" for stacked, geom_point() for the dot plot, geom_tile() for the heatmap.

ANSWER:

Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.DataFrame({
    'Region': ['North','North','North','North',
              'South','South','South','South',
              'East','East','East','East',
              'West','West','West','West'],
    'Quarter': ['Q1','Q2','Q3','Q4',
               'Q1','Q2','Q3','Q4',
               'Q1','Q2','Q3','Q4',
               'Q1','Q2','Q3','Q4'],
    'Sales_000s': [120,135,128,150,
                  95,110,105,130,
                  140,138,145,160,
                  80,85,90,100]
})

pivot = df.pivot(index='Region', columns='Quarter', values='Sales_000s')
pivot.plot(kind='bar', figsize=(8,5))
plt.title("Sales by Region and Quarter (Grouped Bar Chart)")
plt.xlabel("Region")
plt.ylabel("Sales (000s)")
```



```
plt.legend(title="Quarter")
plt.tight_layout()
plt.show()
pivot.plot(kind='bar',
           stacked=True,
           figsize=(8,5))
plt.title("Sales by Region and Quarter (Stacked Bar Chart)")
plt.xlabel("Region")
plt.ylabel("Sales (000s)")
plt.legend(title="Quarter")
plt.tight_layout()
plt.show()
```

```
total_sales = df.groupby('Region')['Sales_000s'].sum().sort_values(ascending=False)
```

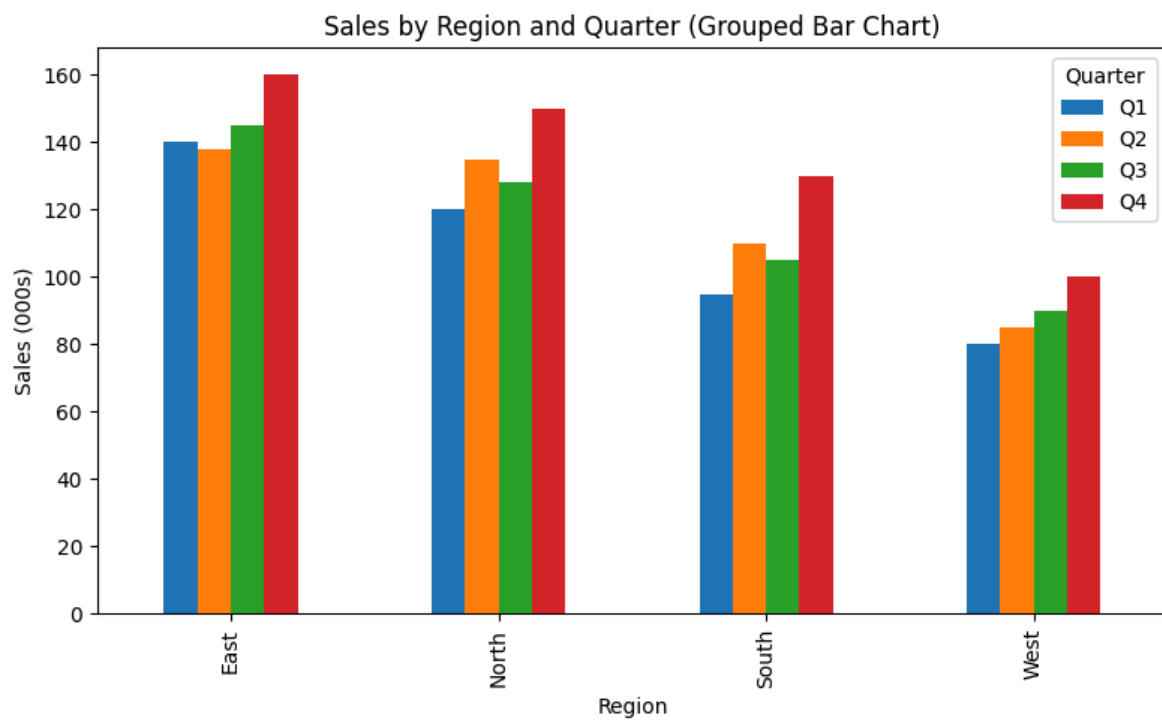
```
plt.figure(figsize=(6,4))
plt.scatter(total_sales.index,
           total_sales.values,
           color='red',
           s=100)
```

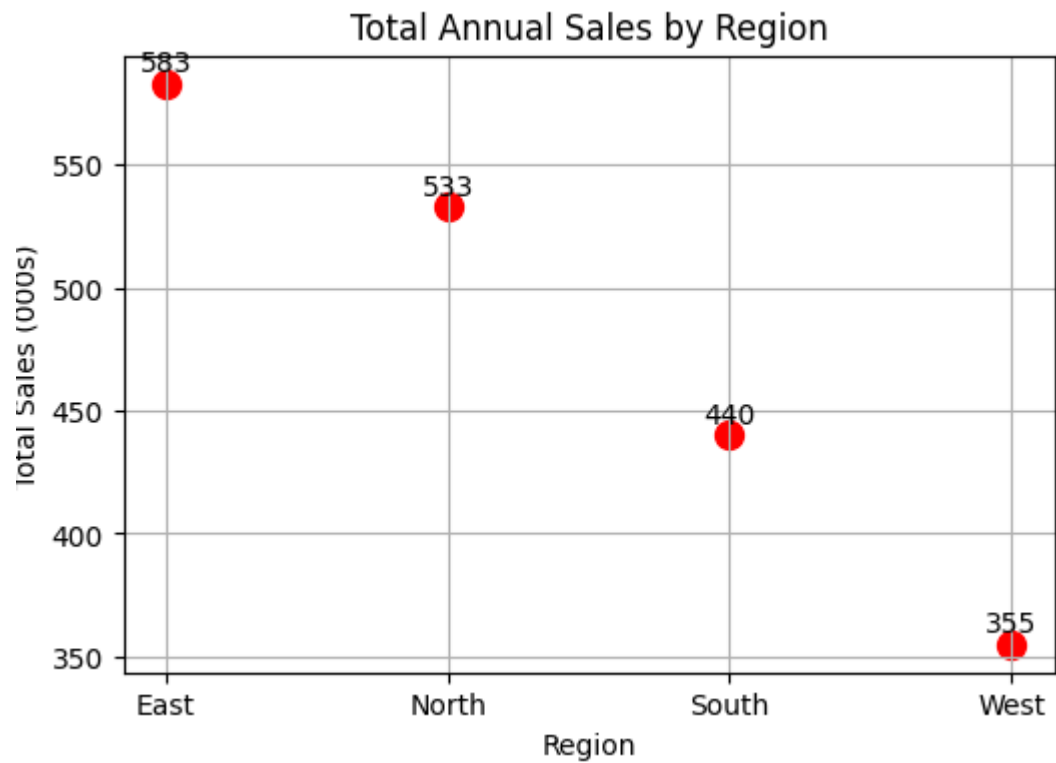
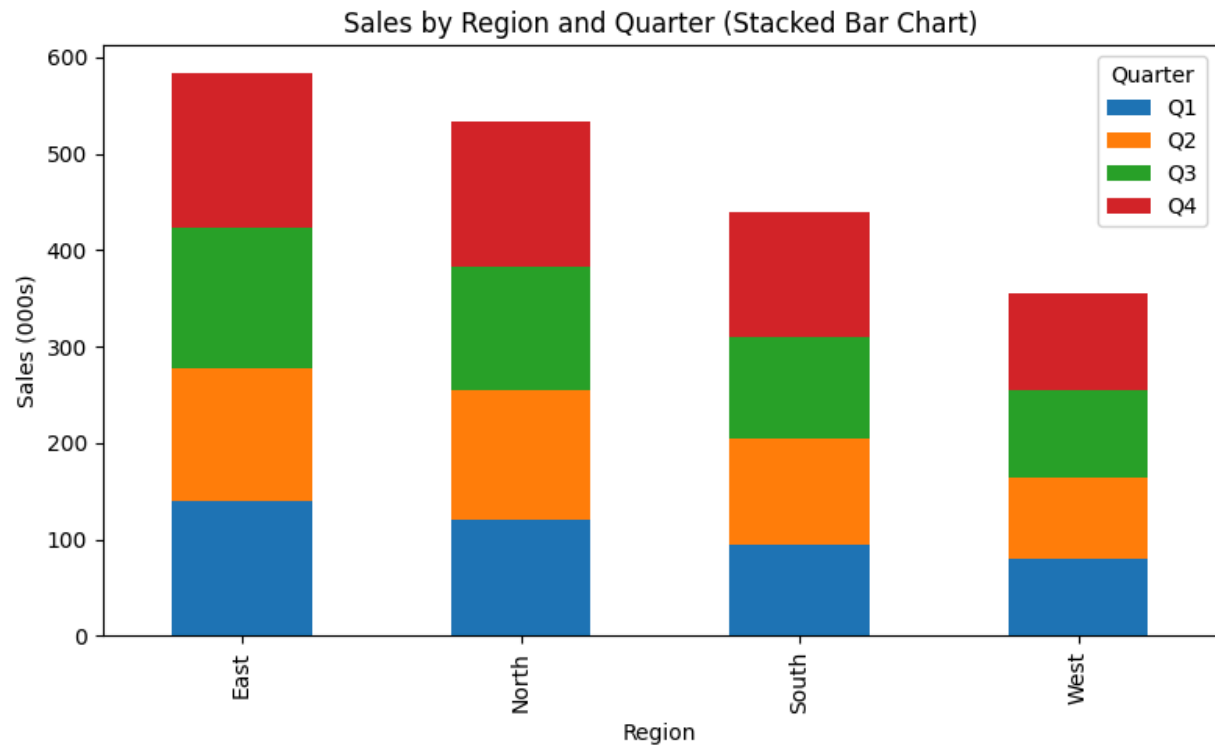
```
for i, value in enumerate(total_sales.values):
    plt.text(i, value+5, str(value), ha='center')
```

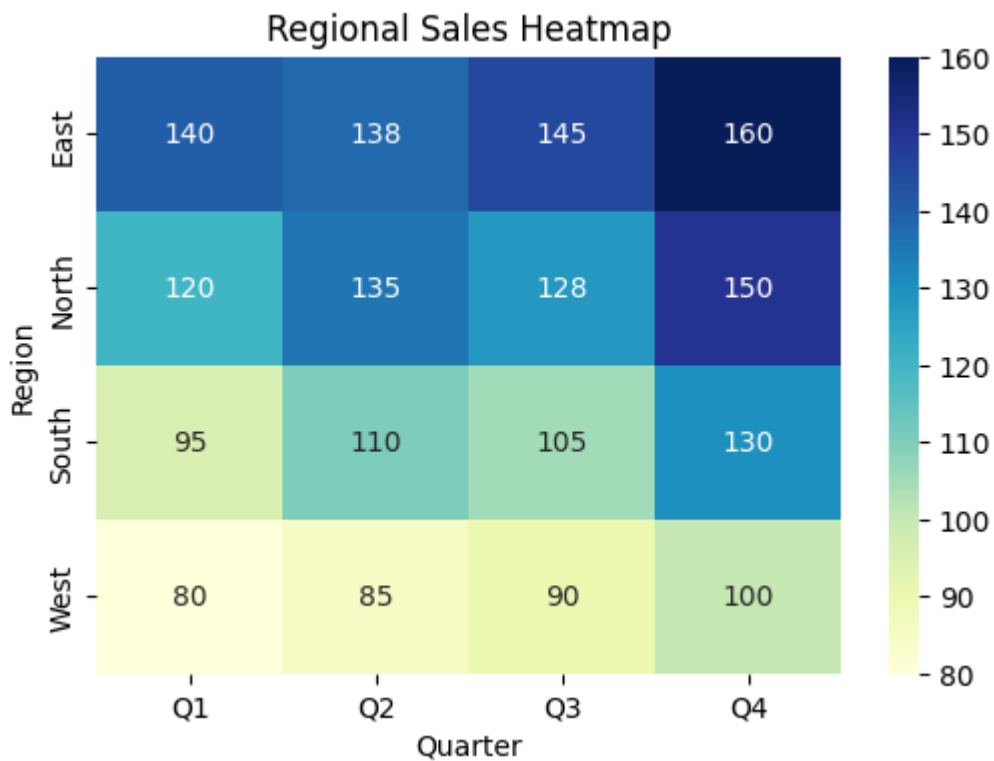
```
plt.title("Total Annual Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales (000s)")
plt.grid(True)
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.heatmap(pivot,
            annot=True,
            cmap='YlGnBu',
            fmt='d')
plt.title("Regional Sales Heatmap")
plt.xlabel("Quarter")
plt.ylabel("Region")
plt.show()
```

Outputs:







R Code

```
library(ggplot2)
library(tidyr)
library(dplyr)
df <- data.frame(
  Region = c("North", "North", "North", "North",
             "South", "South", "South", "South",
             "East", "East", "East", "East",
             "West", "West", "West", "West"),

  Quarter = c("Q1", "Q2", "Q3", "Q4",
              "Q1", "Q2", "Q3", "Q4",
              "Q1", "Q2", "Q3", "Q4",
              "Q1", "Q2", "Q3", "Q4"),
```

```
Sales_000s = c(120,135,128,150,  
              95,110,105,130,  
              140,138,145,160,  
              80,85,90,100)  
)
```

```
ggplot(df,  
       aes(x=Region,  
           y=Sales_000s,  
           fill=Quarter))+
```

```
geom_col(position="dodge")+
```

```
labs(title="Sales by Region and Quarter (Grouped Bar Chart)",  
     x="Region",  
     y="Sales (000s)")+
```

```
theme_minimal()
```

```
ggplot(df,  
       aes(x=Region,  
           y=Sales_000s,  
           fill=Quarter))+
```

```
geom_col(position="stack")+
```

```
labs(title="Sales by Region and Quarter (Stacked Bar Chart)",  
     x="Region",  
     y="Sales (000s)")+
```

```
theme_minimal()
```

```
total_sales <- df %>%
```

```
group_by(Region) %>%
```

```
summarise(TotalSales=sum(Sales_000s)) %>%
```

```
arrange(desc(TotalSales))
```

```
ggplot(total_sales,
```

```
  aes(x=reorder(Region,-TotalSales),
```

```
      y=TotalSales))+
```

```
geom_point(size=4,
```

```
  color="red")+
```

```
geom_text(aes(label=TotalSales),
```

```
  vjust=-1))+
```

```
labs(title="Total Annual Sales by Region",
```

```
  x="Region",
```

```
  y="Total Sales (000s)")+
```

```
theme_minimal()
```

```
ggplot(df,
```

```
  aes(x=Quarter,
```

```
      y=Region,
```

```
      fill=Sales_000s))+
```

```
geom_tile(color="white")+
```

```

geom_text(aes(label=Sales_000s),
          color="black")+

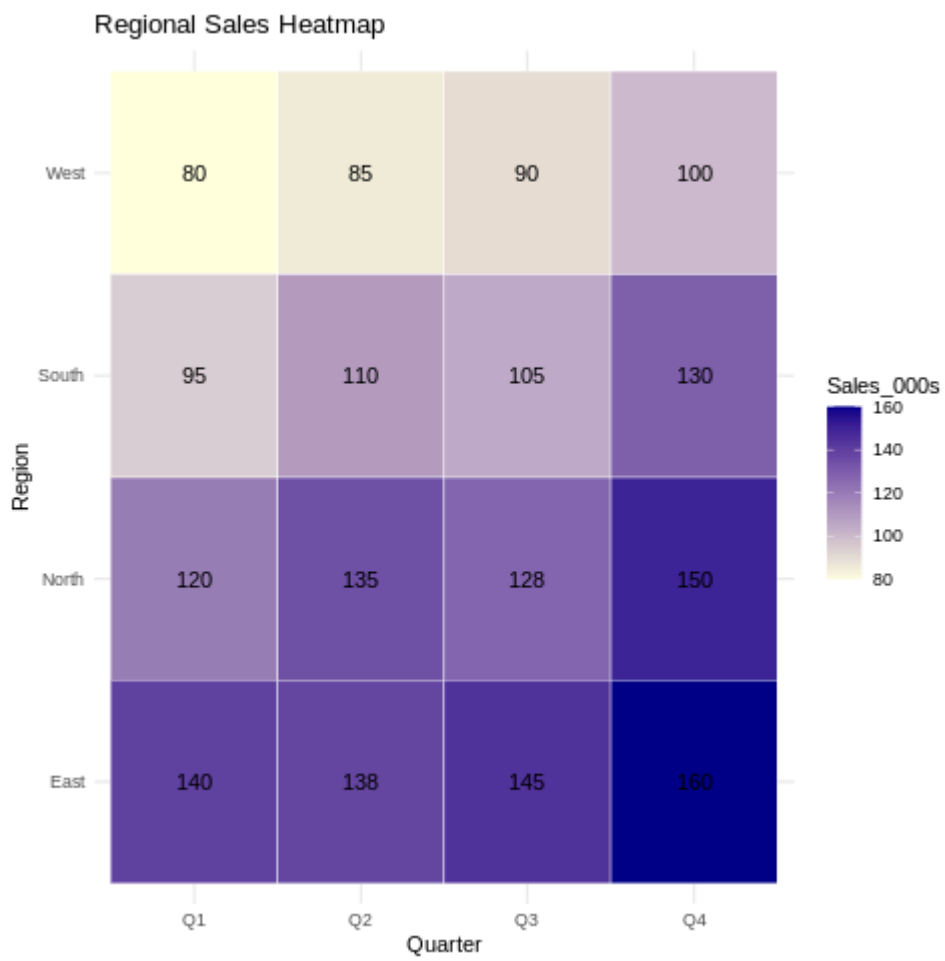
scale_fill_gradient(low="lightyellow",
                   high="darkblue")+

labs(title="Regional Sales Heatmap",
      x="Quarter",
      y="Region")+

theme_minimal()

```

Output:



Question 4

Dataset — generate synthetically for reproducibility:

Class A: 40 exam scores ~ Normal(mean=70, sd=8)

Class B: 40 exam scores ~ Normal(mean=75, sd=5)

Class C: 40 exam scores ~ Normal(mean=65, sd=12)

Class D: 40 exam scores ~ Normal(mean=80, sd=6)

(Set a random seed of 42 in both Python and R so results are consistent across languages.) **Task:**

1. Violin plot: exam score distribution by class (all 4 classes on one plot), with an embedded boxplot or median marker.
2. Ridgeline plot: same 4 classes, overlapping density curves, ordered by mean score.

Python: `numpy.random.normal` to generate data, `seaborn.violinplot` for the violin plot, `joypy.joyplot` (or a stacked `seaborn/matplotlib kdeplot FacetGrid`) for the ridgeline.

R: `rnorm()` to generate data, `ggplot2::geom_violin()` for the violin plot, `ggridges::geom_density_ridges()` for the ridgeline.

ANSWER:

Python Code

```
!pip install joypy
```

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import joypy
```

```
np.random.seed(42)
```

```
class_A = np.random.normal(70, 8, 40)
```

```
class_B = np.random.normal(75, 5, 40)
```



```

class_C = np.random.normal(65, 12, 40)
class_D = np.random.normal(80, 6, 40)

df = pd.DataFrame({
    'Score': np.concatenate([class_A, class_B, class_C, class_D]),
    'Class': ['Class A']*40 + ['Class B']*40 + ['Class C']*40 + ['Class D']*40
})

plt.figure(figsize=(8,5))

sns.violinplot(
    x='Class',
    y='Score',
    data=df,
    inner='box',
    palette='Set2'
)

plt.title("Exam Score Distribution by Class")
plt.xlabel("Class")
plt.ylabel("Exam Score")

plt.show()

order = df.groupby('Class')['Score'].mean().sort_values().index

df['Class'] = pd.Categorical(df['Class'],
                             categories=order,
                             ordered=True)

```

```

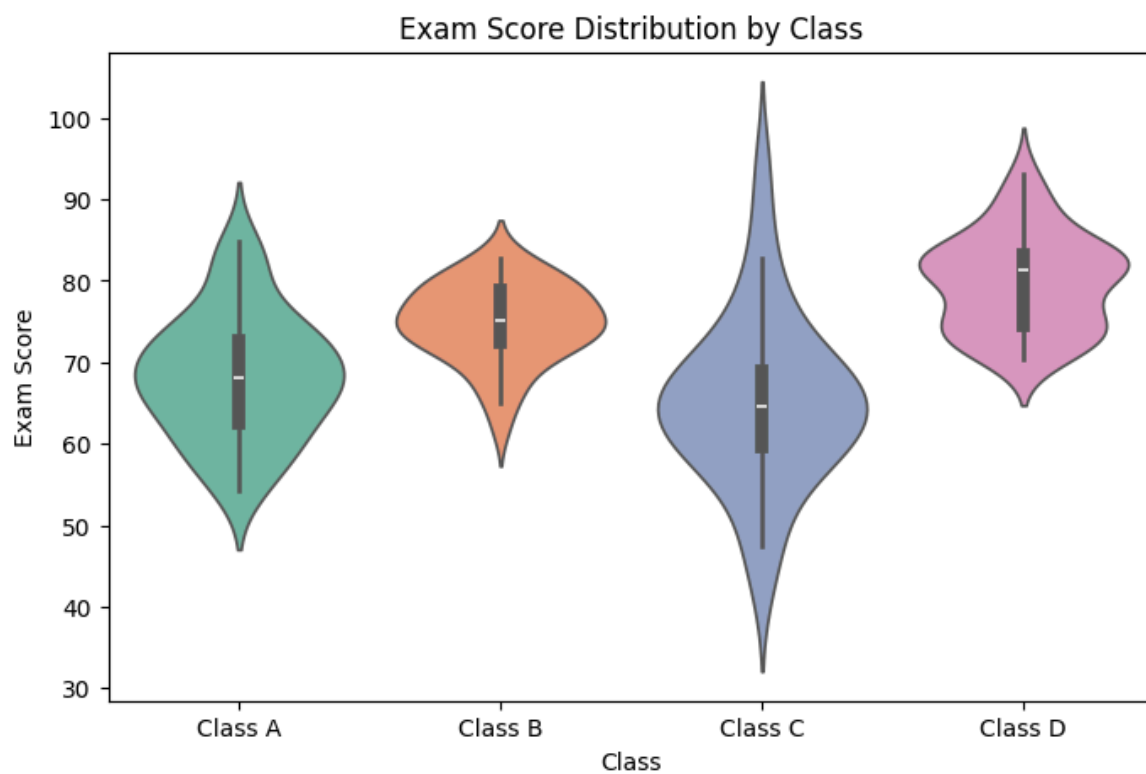
joyppy.joyplot(
    df,
    by='Class',
    column='Score',
    figsize=(8,6),
    colormap=plt.cm.Set2,
    overlap=1
)

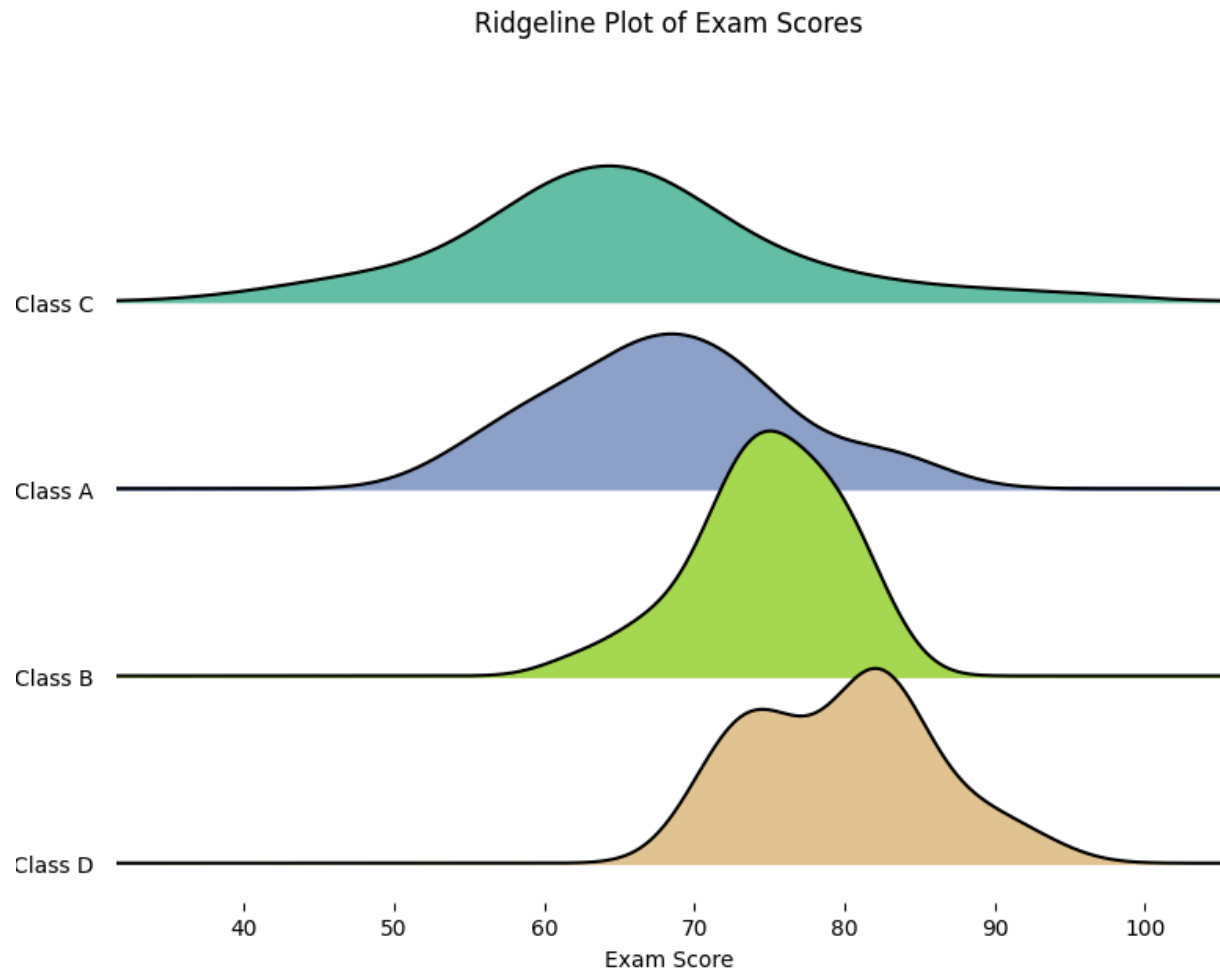
plt.title("Ridgeline Plot of Exam Scores")
plt.xlabel("Exam Score")

plt.show()

```

Output:





R Code

```
install.packages("ggridges")  
# Load libraries  
library(ggplot2)  
library(ggridges)  
  
# Set random seed  
set.seed(42)  
  
# Generate data  
class_A <- rnorm(40, mean = 70, sd = 8)  
class_B <- rnorm(40, mean = 75, sd = 5)
```

```

class_C <- rnorm(40, mean = 65, sd = 12)
class_D <- rnorm(40, mean = 80, sd = 6)

# Create DataFrame
df <- data.frame(
  Score = c(class_A, class_B, class_C, class_D),
  Class = factor(c(rep("Class A",40),
                    rep("Class B",40),
                    rep("Class C",40),
                    rep("Class D",40)))
)

ggplot(df,
  aes(x = Class,
      y = Score,
      fill = Class)) +

  geom_violin(trim = FALSE) +

  geom_boxplot(width = 0.1,
              fill = "white") +

  labs(title = "Exam Score Distribution by Class",
       x = "Class",
       y = "Exam Score") +

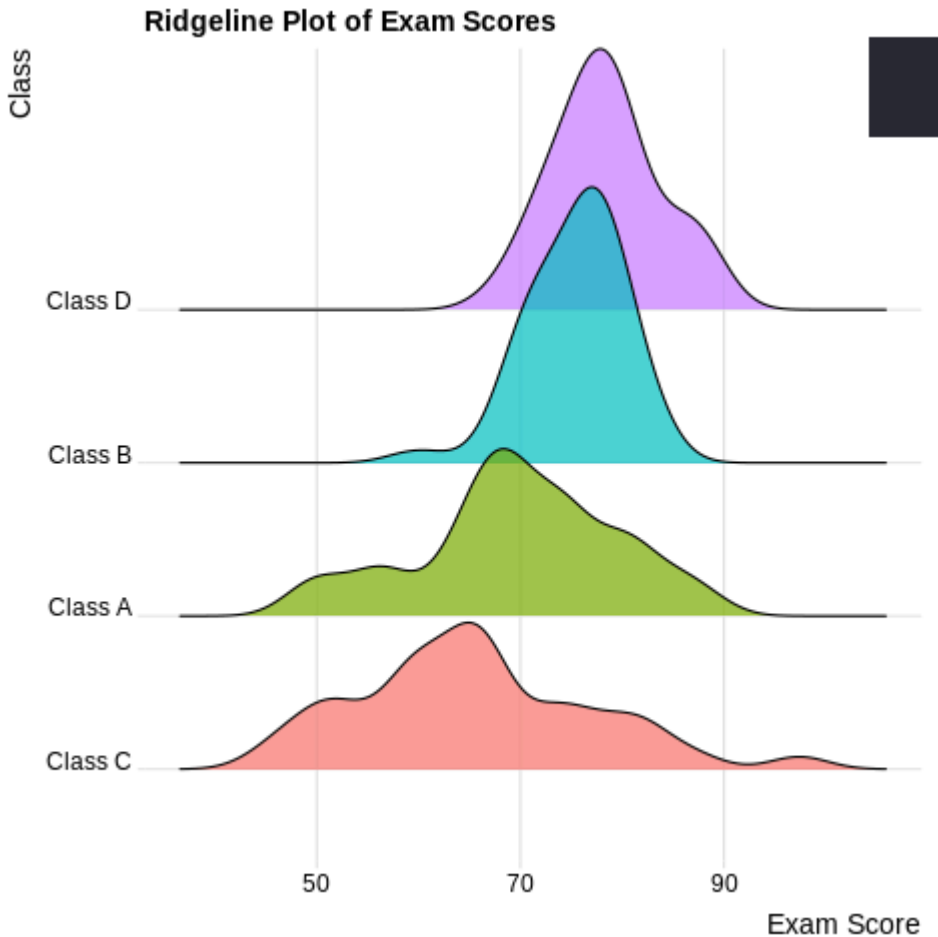
  theme_minimal()

# -----
# 2. Ridgeline Plot

```

```
# -----  
  
# Order classes by mean score  
means <- aggregate(Score ~ Class, df, mean)  
  
df$Class <- factor(df$Class,  
                  levels = means$Class[order(means$Score)])  
  
ggplot(df,  
       aes(x = Score,  
           y = Class,  
           fill = Class)) +  
  
  geom_density_ridges(alpha = 0.7) +  
  
  labs(title = "Ridgeline Plot of Exam Scores",  
       x = "Exam Score",  
       y = "Class") +  
  
  theme_ridges() +  
  
  theme(legend.position = "none")
```

Output:



Question 5

Dataset: Use a built-in dataset available in both languages.

- Python: `seaborn.load_dataset("tips")` → use the `total_bill` column.
- R: the equivalent `tips` dataset (available via the `reshape2` package, or import the same CSV used in Python so results are directly comparable).

Task:

1. Plot a histogram of the chosen continuous variable with 30 bins.
2. Overlay a kernel density estimate (KDE) curve on the same plot.
3. Repeat both, split/colored by the smoker categorical variable, to compare distribution shape across groups. **Python:** `seaborn.histplot(..., kde=True)` or `matplotlib.pyplot.hist + scipy.stats.gaussian_kde`.

R: `ggplot2::geom_histogram()` combined with `geom_density()` (use `after_stat(density)` so scales match), with `fill =` mapped to smoker for the grouped version.

ANSWER:

Python Code

```
# Import libraries
import seaborn as sns
import matplotlib.pyplot as plt

# Load built-in dataset
tips = sns.load_dataset("tips")

plt.figure(figsize=(8,5))

sns.histplot(
    data=tips,
    x="total_bill",
    bins=30,
    kde=True,
    color="skyblue"
)

plt.title("Histogram with KDE of Total Bill")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")

plt.show()

plt.figure(figsize=(8,5))

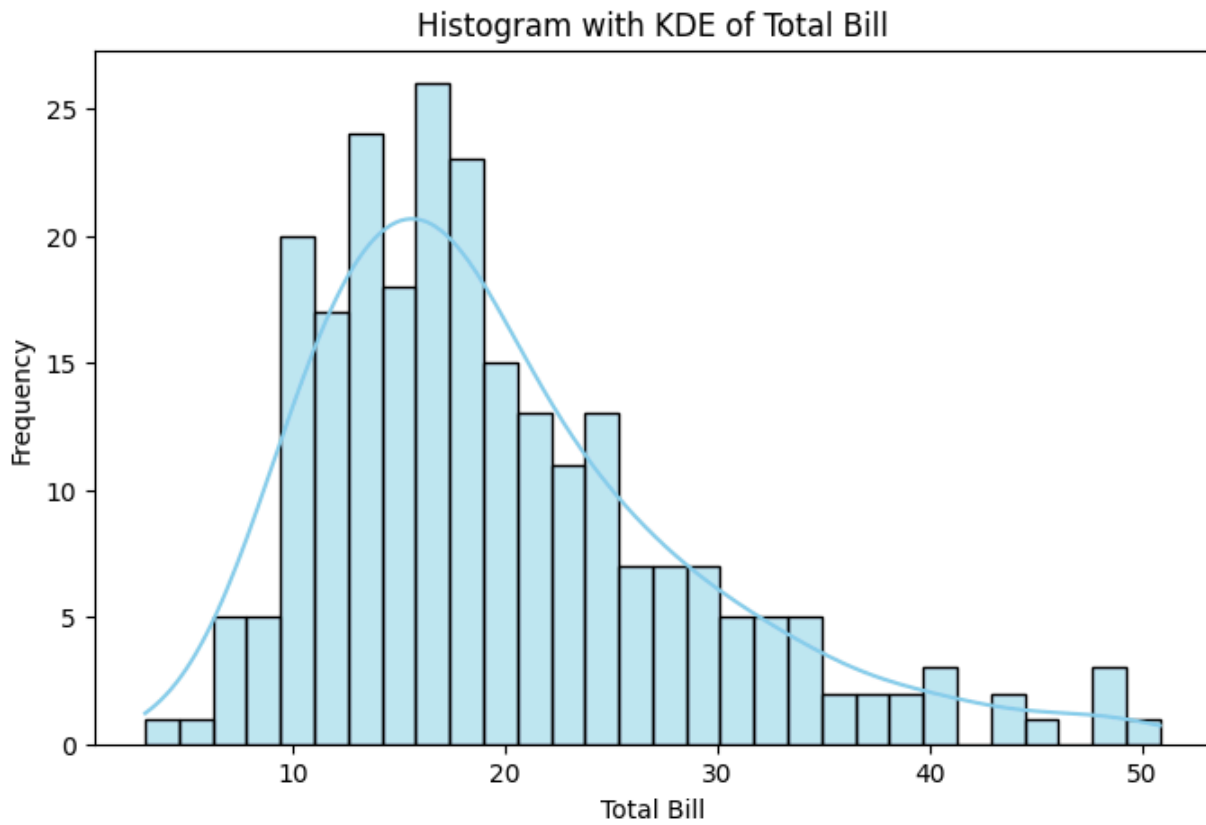
sns.histplot(
    data=tips,
    x="total_bill",
    bins=30,
    hue="smoker",
    kde=True,
    multiple="layer"
)

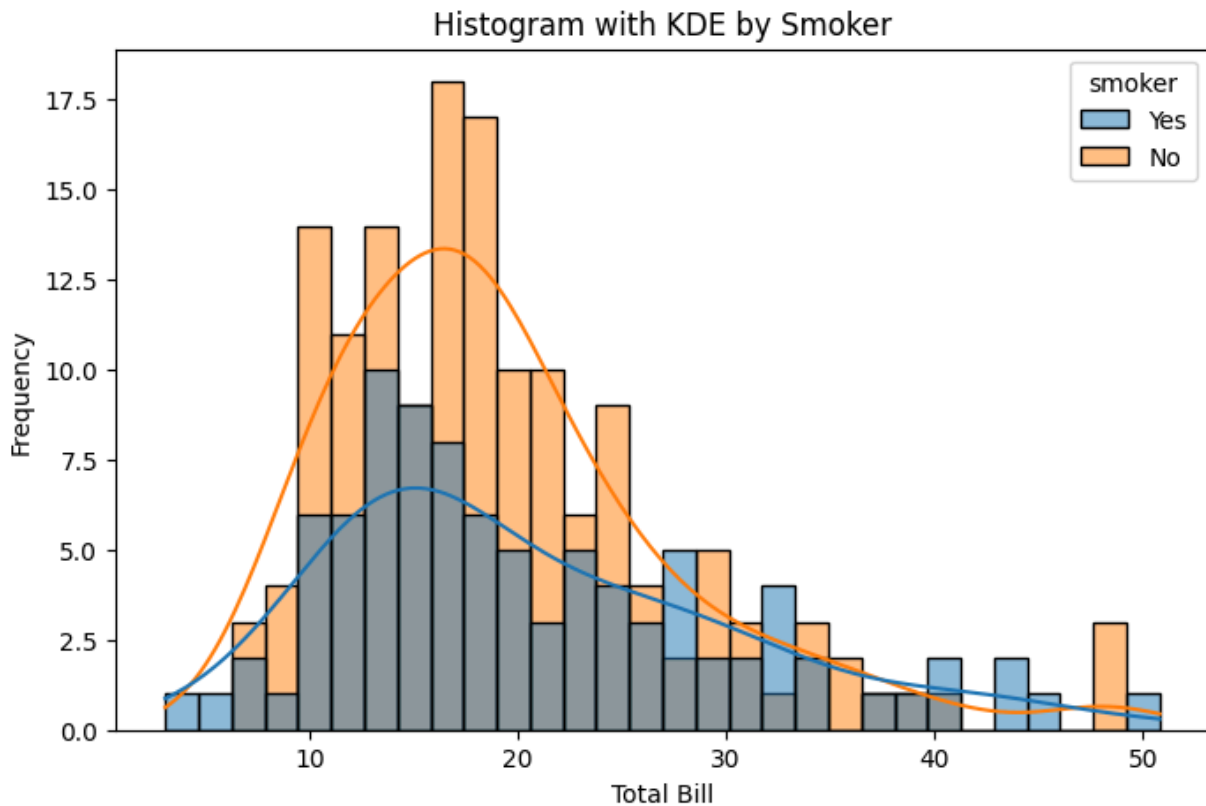
plt.title("Histogram with KDE by Smoker")
```

```
plt.xlabel("Total Bill")  
plt.ylabel("Frequency")
```

```
plt.show()
```

Output:





R Code

```
# Install package if needed
# install.packages("reshape2")
```

```
# Load libraries
library(ggplot2)
library(reshape2)
```

```
# Load dataset
data("tips")
```

```
ggplot(tips,
  aes(x = total_bill)) +

  geom_histogram(aes(y = after_stat(density)),
    bins = 30,
    fill = "skyblue",
```

```

    color = "black") +

geom_density(color = "red",
  linewidth = 1) +

labs(title = "Histogram with Density Curve of Total Bill",
  x = "Total Bill",
  y = "Density") +

theme_minimal()
ggplot(tips,
  aes(x = total_bill,
    fill = smoker,
    color = smoker)) +

geom_histogram(aes(y = after_stat(density)),
  bins = 30,
  alpha = 0.5,
  position = "identity") +

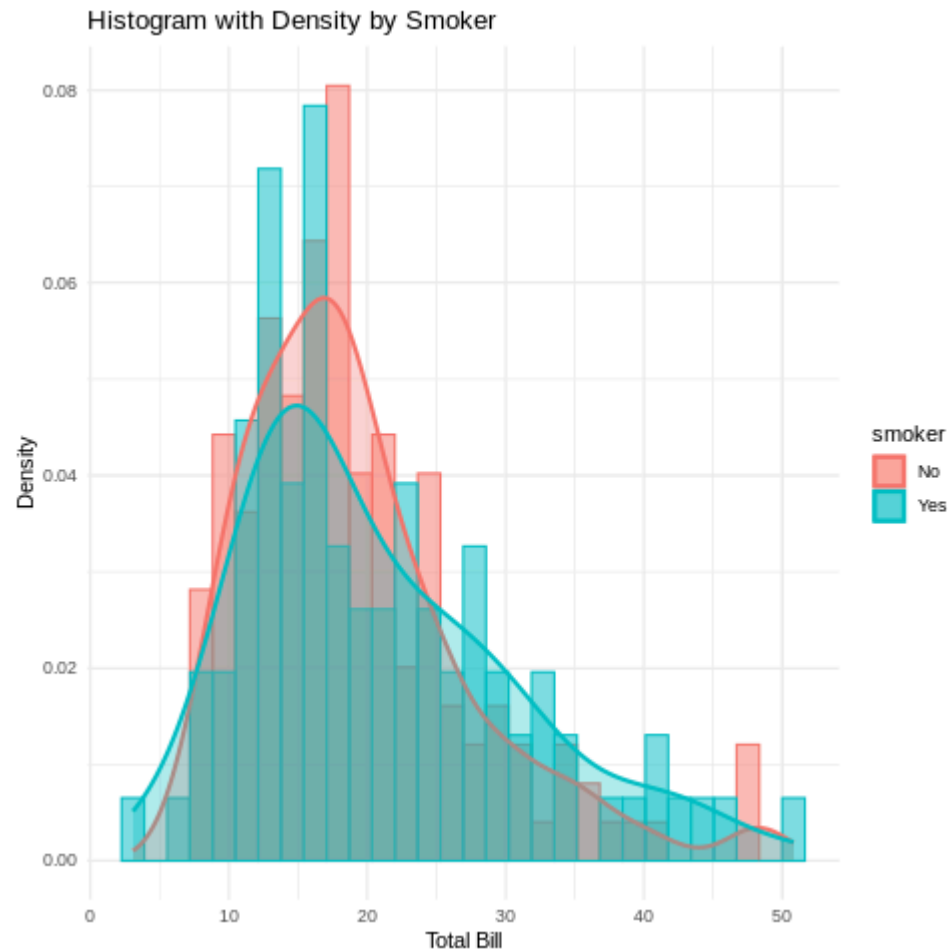
geom_density(alpha = 0.3,
  linewidth = 1) +

labs(title = "Histogram with Density by Smoker",
  x = "Total Bill",
  y = "Density") +

theme_minimal()

```

Output:



Submission Checklist

- Q1: written chart-type justifications (no code)
- Q2: bar plot — Python + R
- Q3: grouped bar, stacked bar, dot plot, heatmap — Python + R
- Q4: violin plot, ridgeline plot — Python + R
- Q5: histogram + density (overall and grouped) — Python + R
- Each chart has a title, axis labels, and a short written interpretation